



Measurement and Sensor Technologies

5. Outline of Sensor Technologies

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Outline

5.1

Basic Concepts of Sensors

5.2

Sensor Calibration

5.3

Principles of Instrument Selection

5.4

Applications and Development of Sensor



Outline

5.1

Basic Concepts of Sensors

- 5.1.1 Definition and Composition of Sensors
- 5.1.2 Function of Sensors
- 5.1.3 Classification of Sensors
- 5.1.4 Basic Characteristics of Sensors

5.1 Basic Concepts of Sensors

- Sensor is omnipresent (无所不在的) in the industrial and daily life.



Camera image sensor



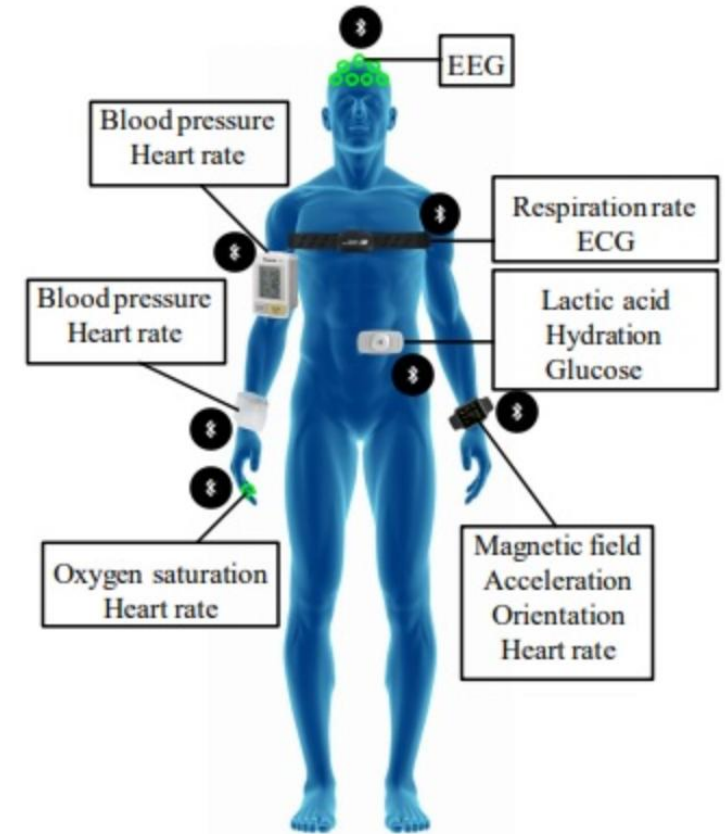
Air quality detector & sensor



Magnetic proximity sensor
for food processing industry



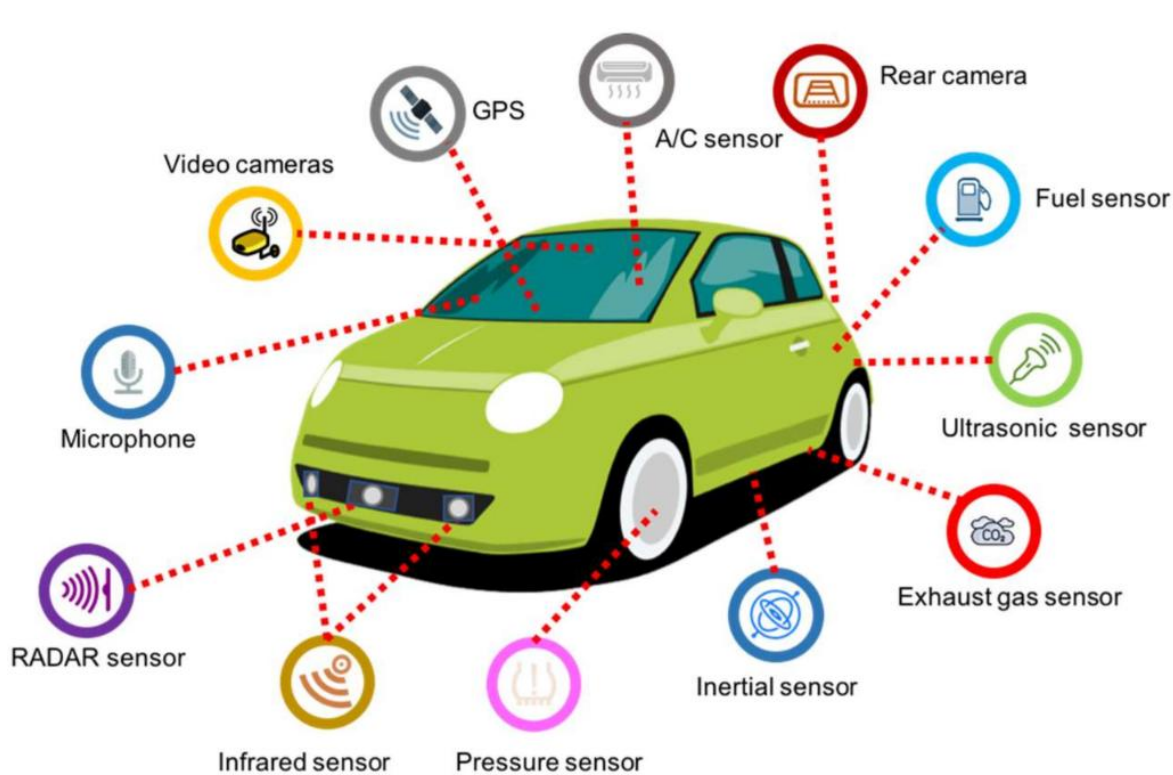
Suppression sensor
for food safety



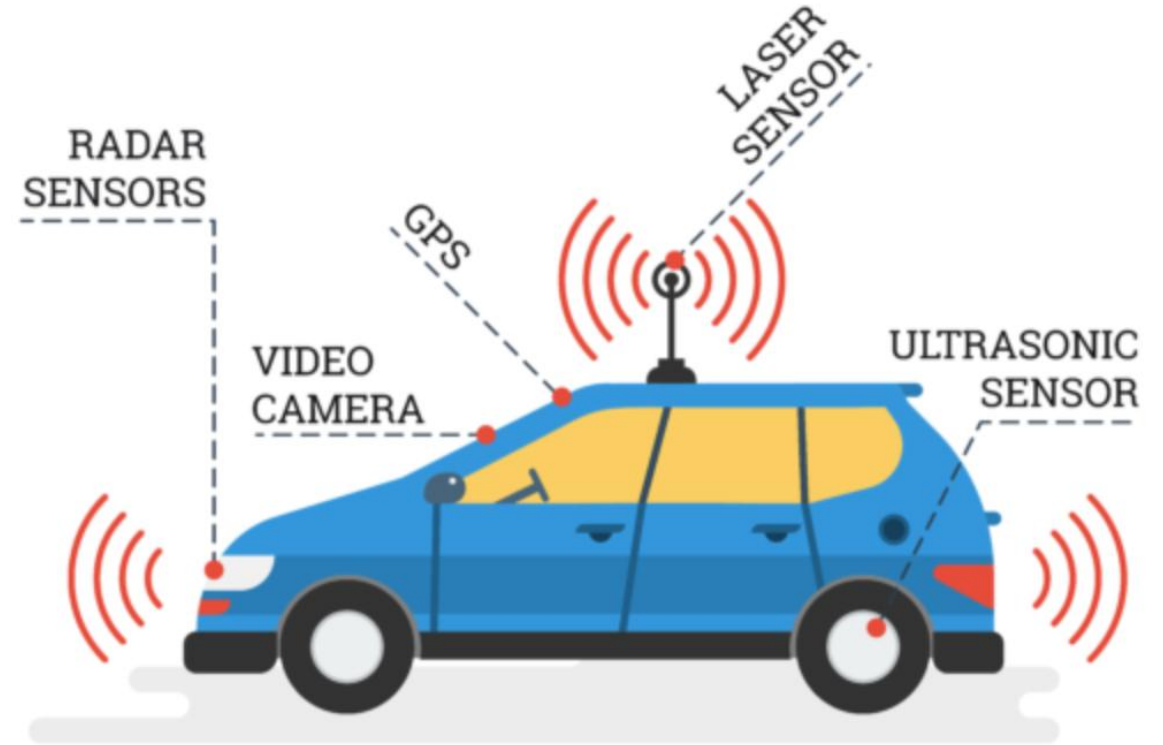
Medical treatment
sensors

5.1 Basic Concepts of Sensors

- Sensor is omnipresent in the industrial and daily life.



Different types of vehicle sensors

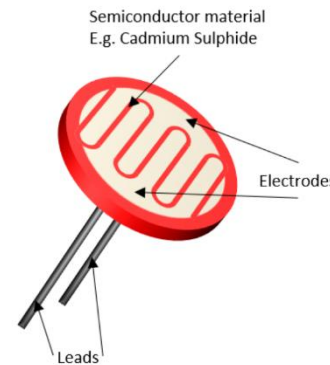
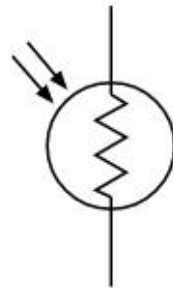
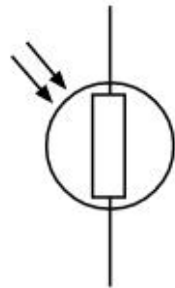


Self-driving sensors

5.1.1 Definition and Composition of Sensors

■ **Definition:** A device which provides an electrical signal output with respect to a specific physical quantity

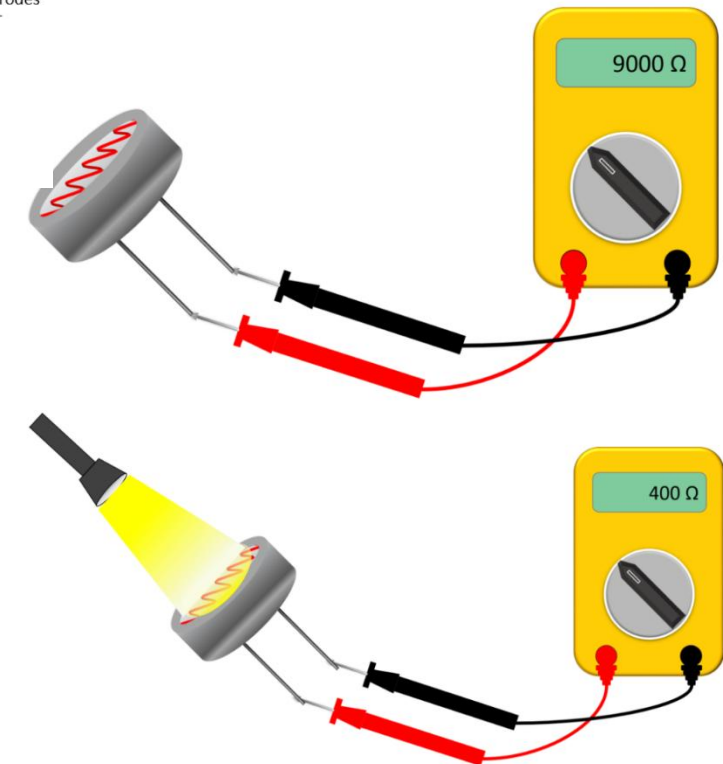
Example 1: Light Dependent Resistor (LDR).



Theory: Resistance varies with the intensity of the light. When light strikes on it, the free electrons gain enough energy to move through the material. Hence, the resistance of a semiconductor material varies

Applications:

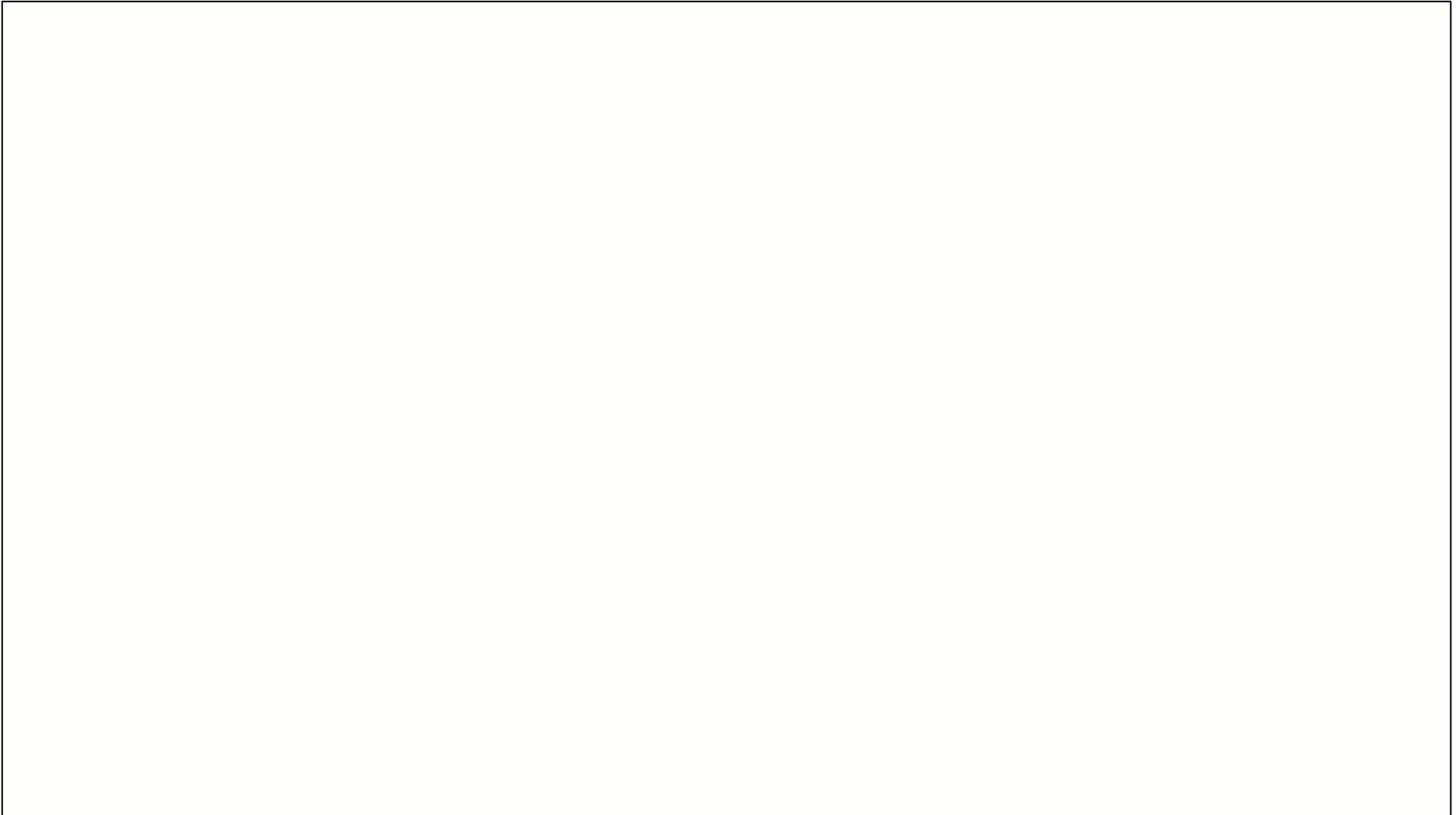
1. Simple Automatic Room Lights Control
2. Automatic Street Lights ON or OFF





5.1.1 Definition and Composition of Sensors

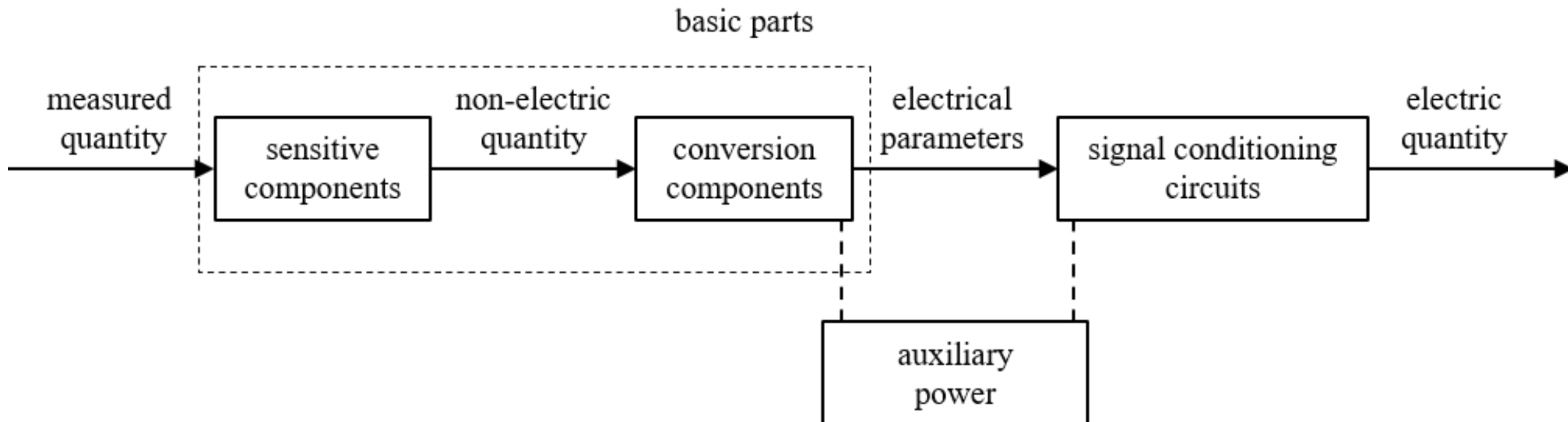
Example 2: Thermopile vs. Thermographic





5.1.1 Definition and Composition of Sensors

- **Composition:** Sensitive components, Conversion components, and Signal conditioning circuits (Sometimes an auxiliary power supply is needed to provide conversion energy)

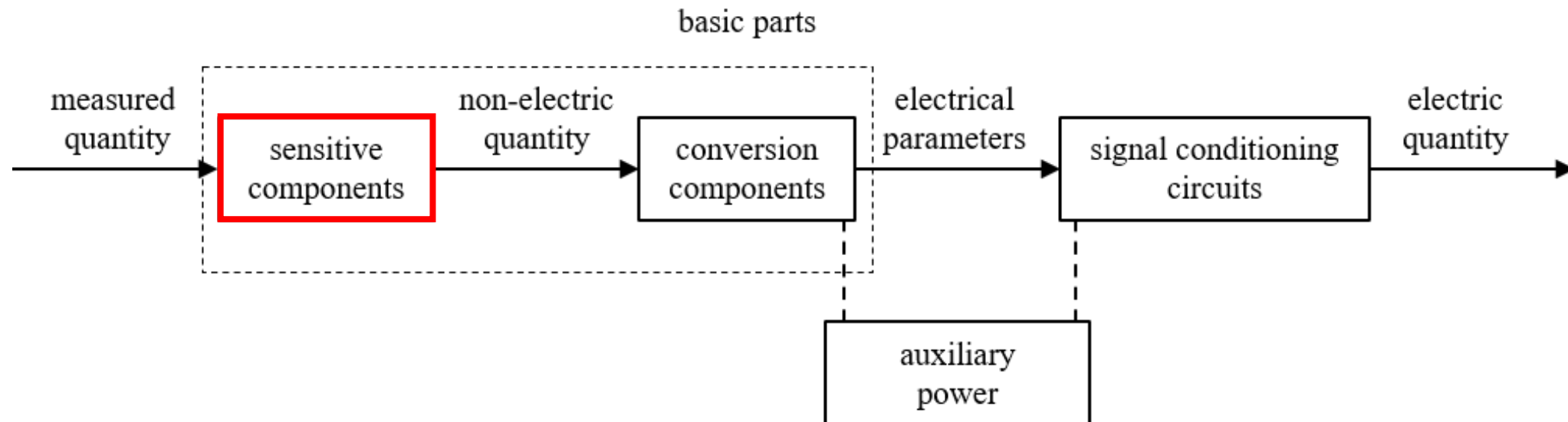




5.1.1 Definition and Composition of Sensors

➤ Sensitive Components (Pre-converter)

- The device that can complete the pre-conversion, also known as the pre-converter
- Note that, not all the non-electric quantity can be directly converted into electric quantity by the existing means

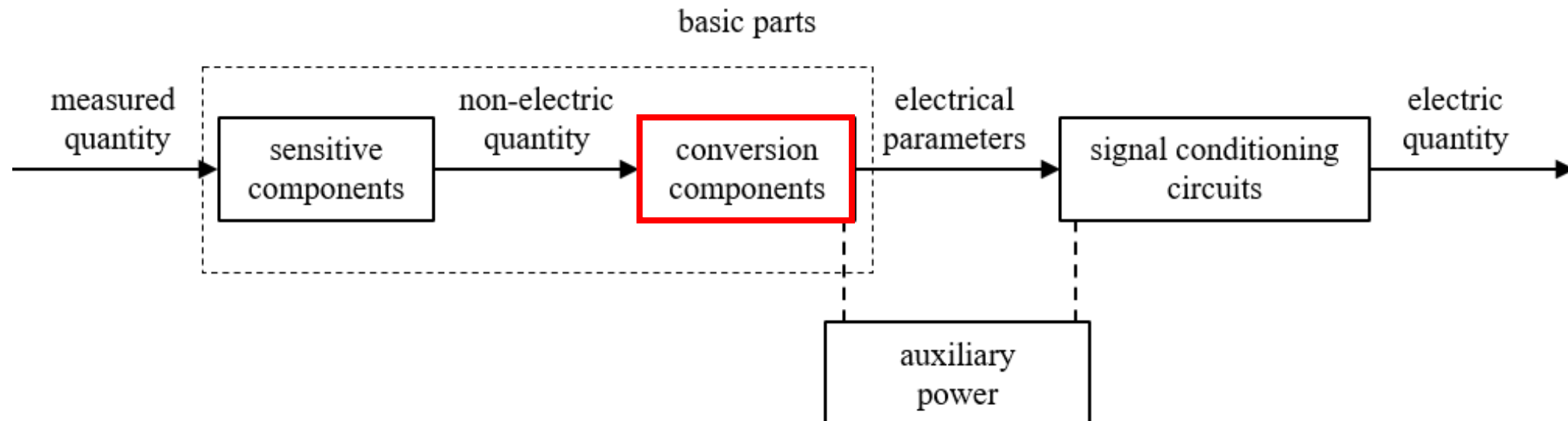




5.1.1 Definition and Composition of Sensors

➤ Conversion Components

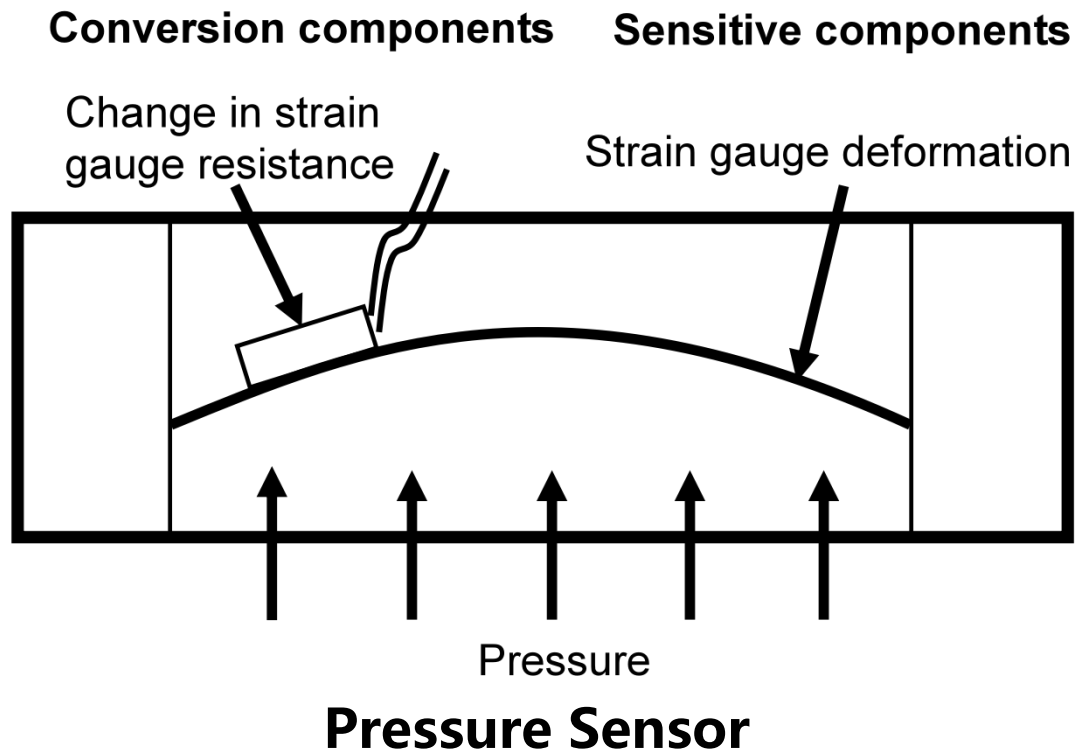
- The components that convert the output of sensitive components into electrical parameters that are easy to transmit and process (such as voltage, current, resistance, inductance, capacitance, etc.)



5.1.1 Definition and Composition of Sensors

➤ Conversion Components

- Not all sensors have sensitive and conversion elements, such as piezoelectric elements, optoelectronic devices, thermocouples, etc., which can directly convert the sensed measurement into electrical signal output



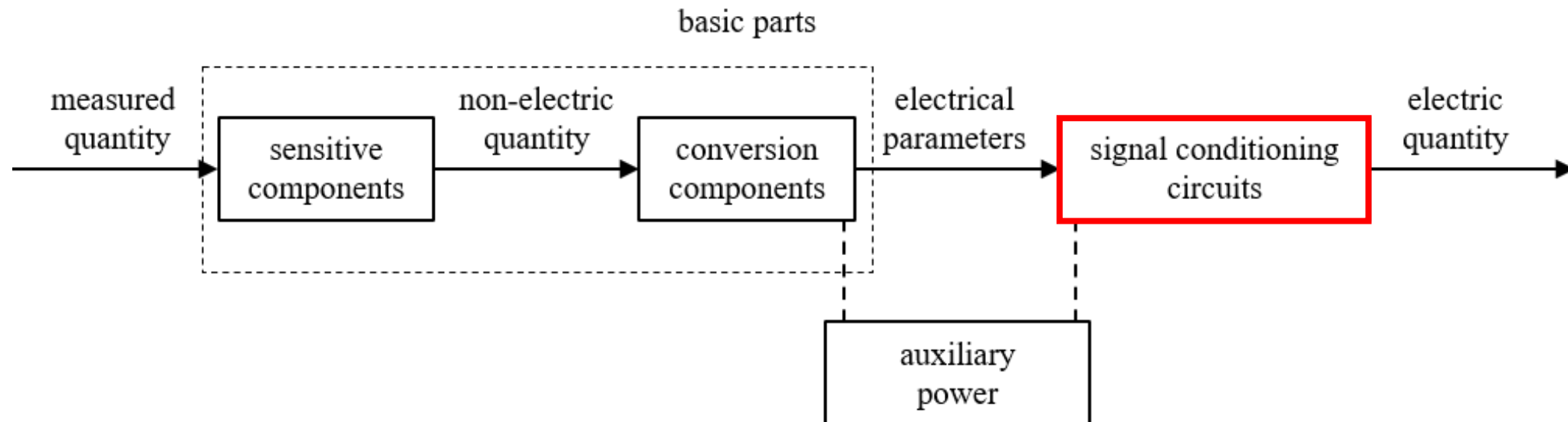
The strain resistance attached to the matrix material changes with the mechanical deformation



5.1.1 Definition and Composition of Sensors

➤ Signal Conditioning and Conversion Circuits

- To amplify and process the output electrical signal of the conversion component, and further convert into the form of easy transmission and processing



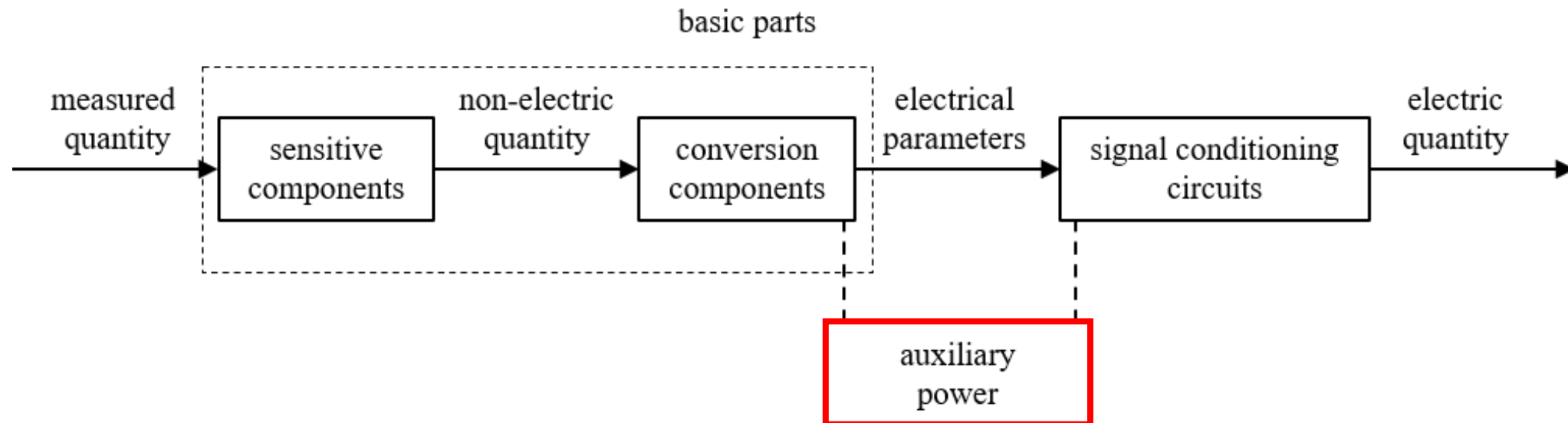


5.1.1 Definition and Composition of Sensors

➤ Auxiliary Power

- To provide energy to the sensor

{ Active sensor: without external power
Passive sensor: with external power

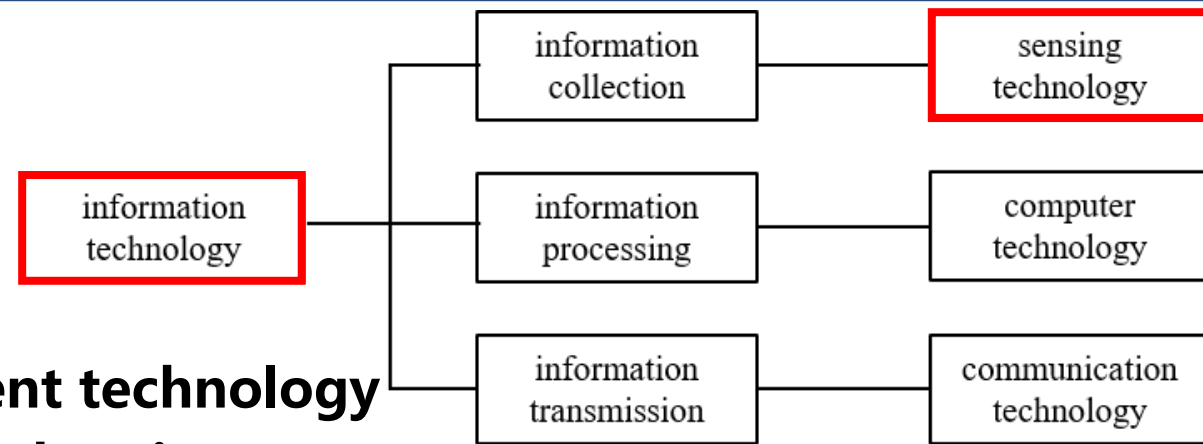




5.1.2 Function of Sensors

➤ Sensor Technology

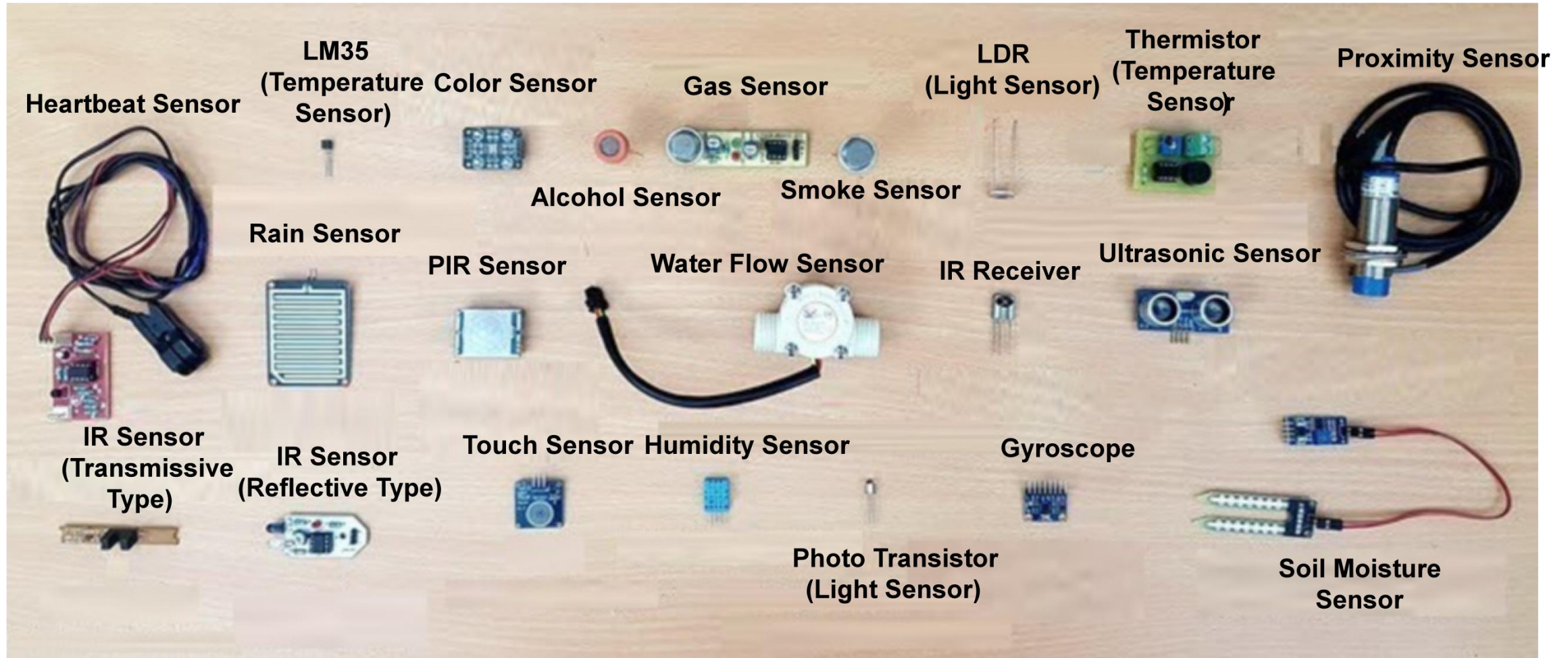
- **Main link of modern control and measurement technology**
- **Photosensitive sensor-vision, acoustic sensor-hearing, gas sensor-smell, chemical sensor-taste, pressure sensitive, temperature sensitive, fluid sensor-touch**
- **Information is acquired through the senses (sensors), commands are sent by the brain (computer), transmitted by the meridians (communication technology), and then calculated and controlled by the computer**
- **Internet of Things (IoT): Perceptual layer, Network layer and Application layer**
- **Perceptual layer relies on Sensor Technology**





5.1.3 Classification of Sensors

➤ Sensor Types





5.1.3 Classification of Sensors

➤ Classification Method

- **External power supply:** Active sensor & Passive sensor
- **Selected reference:** Absolute & Relative sensor
- **Working mechanism of sensors:** physical, chemical and biological
- **Principle of sensor transformation:** resistance type, inductance type, capacitance type, piezoresistive type, piezoelectric type, photoelectric type, magnetic sensitive type, laser, ultrasonic and other sensors
- **Application scope and object**
- **Energy relationship**
-



5.1.3 Classification of Sensors

➤ Classification Method

- Types of stimulus

Stimulus	Sensors
Acoustic	Wave amplitude, phase, polarization; Spectrum; Wave velocity; Other
Biological	Biomass (types, concentration, states); Other
Chemical	Components (identities, concentration, states); Other
Electric	Charge, current; Potential, voltage; Electric field (amplitude, phase, polarization, spectrum); Conductivity; Permittivity; Other
Magnetic	Magnetic field (amplitude, phase, polarization, spectrum); Magnetic flux; Permeability; Other
Optical	Wave amplitude, phase, polarization, spectrum; Wave velocity; Refractive index; Emissivity, reflectivity, absorption; Other
Mechanical	Position (linear, angular); Acceleration; Force; Stress, pressure; Strain; Mass, density; Moment, torque; Speed of flow, rate of mass transport; Shape, roughness, orientation; Stiffness, compliance; Viscosity; Crystallinity, structural integrity; Other
Radiation	Type; Energy; Intensity; Other
Thermal	Temperature; Flux; Specific heat; Thermal conductivity; Other



5.1.4 Basic Characteristics of Sensors

➤ Basic Characteristics of Sensors

- Static characteristics: linearity, sensitivity, hysteresis, repeatability, drift, resolution, threshold, stability, precision ...
- Dynamic characteristics : natural frequency, damping ratio, frequency characteristics, time constant, rise time, response time, overshoot, steady-state error ...

➤ Basic Characteristics of Elastic Sensing Elements

- Elastic deformation to realize measurement: Elastic beam, column, cylinder, diaphragm, membrane box, spring tube, bellows
- Stiffness, Sensitivity, Elastic Hysteresis, Elastic Aftereffect and Creep, Temperature Characteristics, Natural Frequency



5.1.4 Basic Characteristics of Sensors

➤ Stiffness

- Ability of elastic sensitive element to resist deformation under external force

$$k = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta F}{\Delta x} \right) = \frac{dF}{dx}$$

F is the external force acting on the elastic sensing element, x is the deformation of the component under the action of external force.

➤ Sensitivity

- Size of deformation generated by unit external force
- Reciprocal of stiffness

$$s = \frac{1}{k} = \frac{dx}{dF}$$

linear characteristics

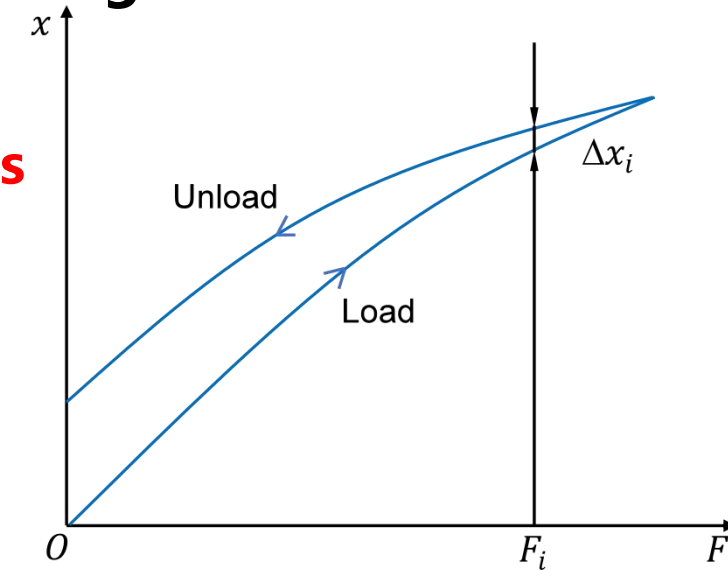


5.1.4 Basic Characteristics of Sensors

➤ Elastic Hysteresis

- Forward and reverse travel curves of the elastic sensing element do not coincide during loading and unloading within the elastic deformation range
- Internal friction between the molecules of the elastic sensing element
- Hysteresis error of the sensor

Small elastic hysteresis



➤ Elastic Aftereffect and Creep

- Failure to recover immediately after the removal of the load but the need for some time to gradually restore to its original state → **repeatability error**
- When the applied load remains constant, the elastic element continues to deform slowly over a long period of time → **stability**



5.1.4 Basic Characteristics of Sensors

➤ Temperature Characteristics

- Variation of ambient temperature → linear thermal expansion

$l = l_0[1 + \beta_l(t - t_0)]$ expansion coefficient β_l , l_0 is used to represent the length when the temperature is t_0

$E = E_0[1 + \beta_E(t - t_0)]$ temperature coefficient of the elastic modulus β_E (negative value). E_0 represents the elastic modulus when the temperature is t_0

- Change of the **geometric size** and **elastic modulus** of the elastic sensing element with the temperature will inevitably lead to the **measurement error**

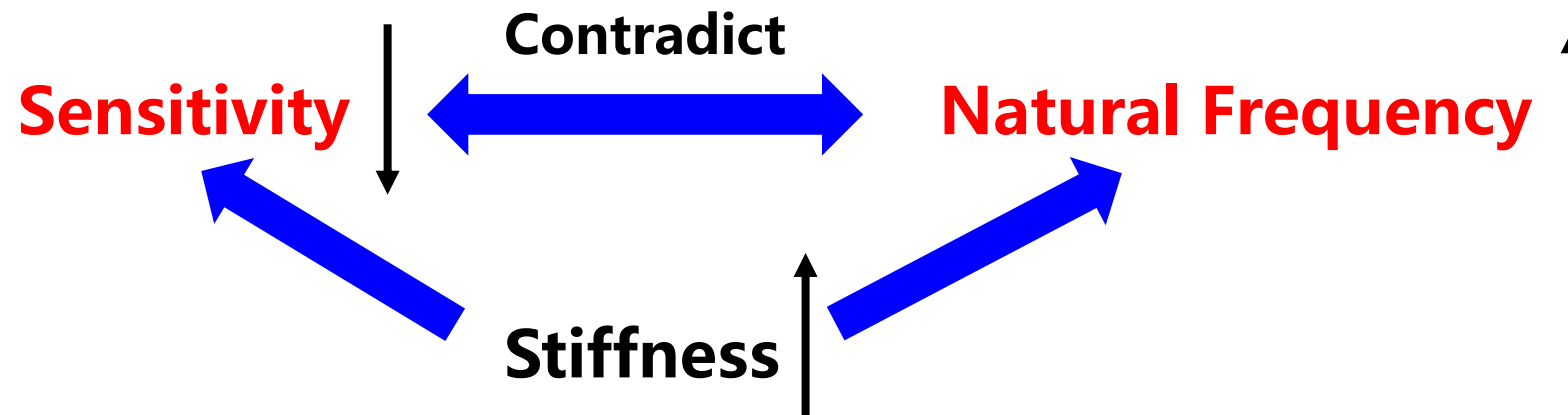


5.1.4 Basic Characteristics of Sensors

➤ Natural Frequency

- Higher Natural Frequency → Better Dynamic characteristics
- Fundamental frequency (基频) : first-order natural frequency

Estimation $\omega_n = \sqrt{\frac{k_e}{m_e}}$ k_e represents equivalent stiffness of elastic sensing element; m_e represents the equivalent vibration quality.





Outline

5.2

Sensor Calibration

- **5.2.1 Fundamental Concepts**
- **5.2.2 Basic Methods**
- **5.2.3 Calibration System**



5.2 Sensor Calibration

- The new sensor needs to be calibrated by known standards or standard instruments with a higher accuracy → **to establish the relationship between output and input and determine its performance**
- The accuracy of sensor will change with time and temperature, and there will be a certain zero drift after a long time, so it needs to be re-calibrated. **Proper sensor calibration will minimize uncertainty of measurements and ensure both precision and consistency**



5.2.1 Fundamental Concepts

➤ Sensor Calibration

- A **process** in which the known non-electric quantity (such as standard pressure, acceleration, displacement) generated by a certain level of instrument and equipment is input into the sensor to be calibrated, and the output of the sensor is measured to obtain its dynamic and static characteristics
- **Static calibration**: sensitivity, linearity, hysteresis, repeatability and accuracy
- **Dynamic calibration**: time constant, natural frequency, damping ratio

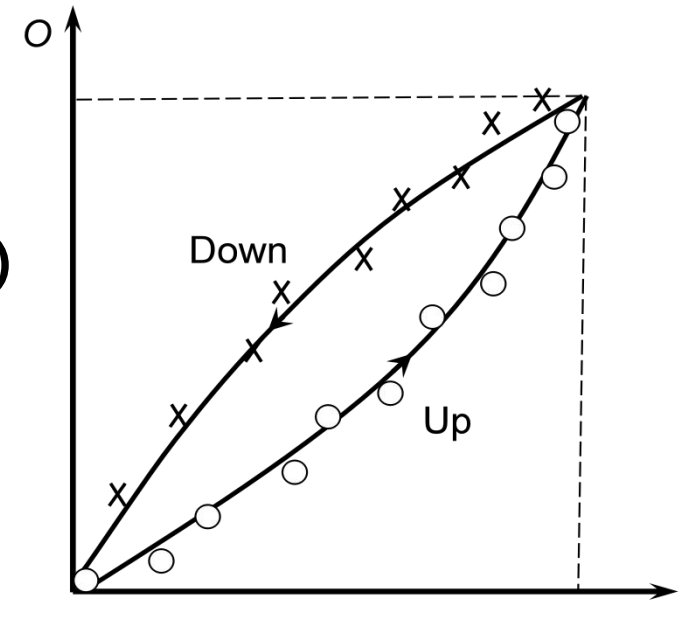
➤ Static Calibration

Conditions

- Static standard conditions (no acceleration, vibration, shock)
- Room temperature $20 \pm 5^\circ\text{C}$
- Relative humidity $< 85\%$
- Atmospheric pressure $100 \pm 7\text{kPa}$

Process

- Standard instrument provides known and invariant input values
- Equally divided n input points
- One input is measured repeatedly m times and each measurement process is repeated for “ups” and “downs”
- Plot curves or calculate the regression equation based on $2m$ groups of data





5.2.2 Basic Methods

➤ Static Calibration

Standard instruments

- **Force calibration** standard instrument (such as force weights, tension and compression dynamometers),
- **Pressure calibration** standard instrument (such as piston pressure gauges, mercury pressure gauges, Maxwell vacuum gauges)
- **Displacement calibration** standard instrument (such as gauge block, ruler)
- **Temperature calibration** standard instrument (such as platinum resistance thermometer)
- **Strain calibration** standard instrument (such as alloy steel with Poisson's ratio of 0.285)



5.2.2 Basic Methods

➤ Dynamic Calibration

Condition and Methods

- Instantaneous physical input varies as a function of time
- Standard actuating signals (sine signals, step signals, etc.)
- Experimental methods to measure the output and input characteristic curves (**Frequency response method** and **Step response method**)

Standard instruments

- Vibration exciter (such as electromagnetic vibration tables, low-frequency rotary tables, mechanical vibration tables, etc.)
- Shock tubes
- Periodic and non-periodic function pressure generators (dynamic calibration of acceleration, velocity, displacement, force, pressure)

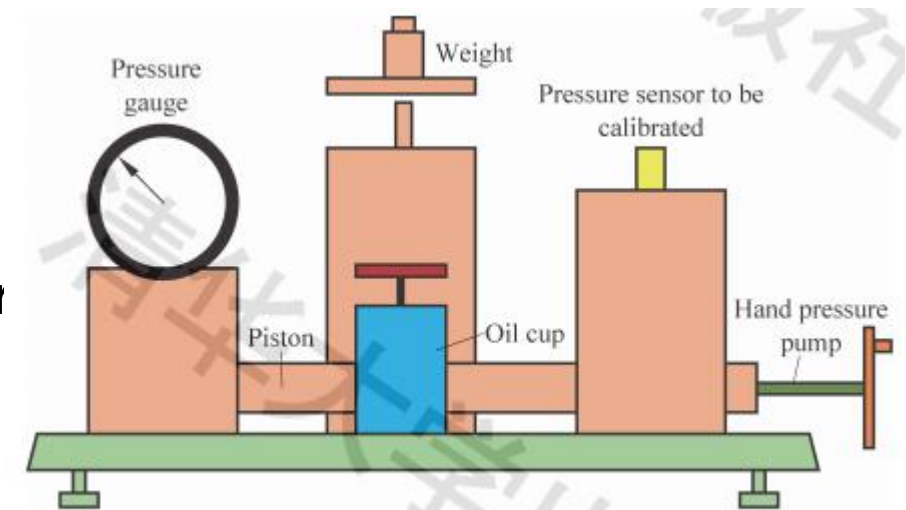
5.2.3 Calibration System

➤ Composition

- Standard generator of non-electric quantity
- Standard test system for non-electric quantity
- Signal detection equipment attached to sensor

Example: Pressure Gauge Calibration

1. Connect pressure gauge to the dead weight tester (DWT) with zero reading
2. Select weight and place it on the vertical piston
3. Adjust the screw pump to ensure the weight and piston freely float on the oil
4. Spin the vertical piston to ensure the free floating
5. Wait for a few moments for system stabilizing, then read record the gauge reading and weight
6. Repeat step 2 to 5 for increasing weights until the full range is applied to the gauge, then decreasing weights until gauge zero reading. Calculate the error at each point

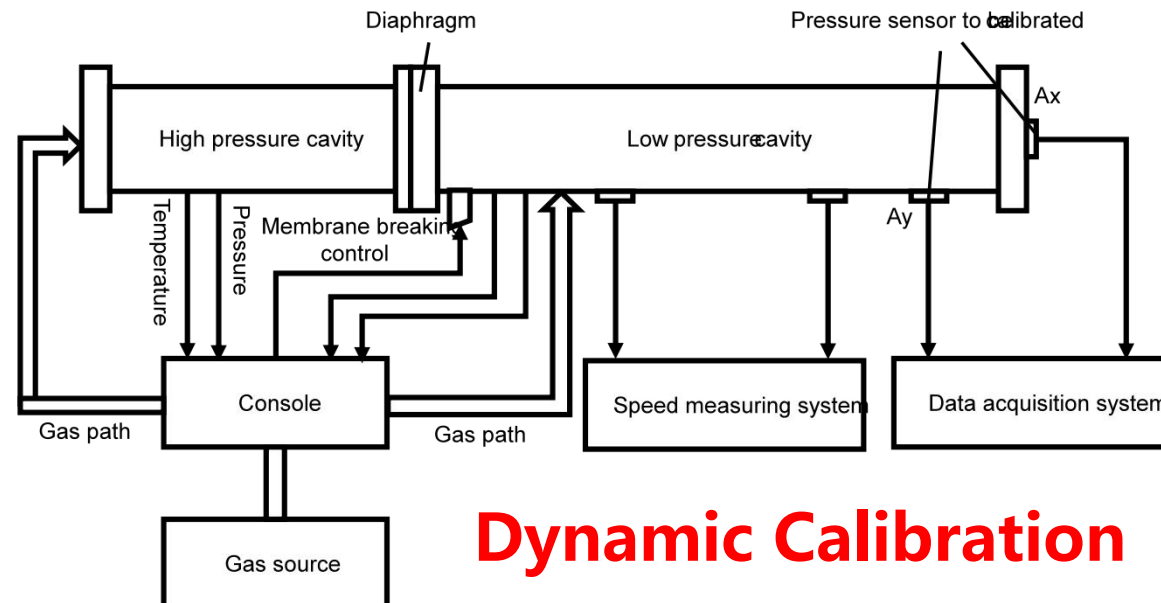




5.2.3 Calibration System

Example 2: Pressure Gauge Calibration with shock tube

- The basic principle of shock tube calibrating the dynamic characteristics is that diaphragm breaks when the pressure difference between high-pressure cavity and low-pressure cavity reaches a certain value, high pressure gas expands rapidly into low pressure cavity, a step wave is formed to stimulate the pressure sensor. The step response of sensor is recorded by the data acquisition system, and the natural frequency and damping ratio are determined according to the step response curve.





Outline

5.3

Principles of Instrument Selection

- **5.3.1 Characteristics of Sensors**
- **5.3.2 Environment of Measurement**
- **5.3.3 Cost and Maintenance**



5.3 Principles of Instrument Selection

How to choose proper sensors for your system?



5.3.1 Characteristics of Sensors

➤ Sensor Characteristics

- **Range, Sensitivity, Precision-related index and Dynamic performance**

Range

- Statement of the limits in which it can function effectively. Lowest and highest input values which it is capable of measuring
- Range and Overload capacity
- Estimate the quantity, avoid exceeding the maximum range of the sensor

Sensitivity

- Sensitivity is the relationship under fixed conditions between a change in the output of a device to the change in input
- Higher sensitivity → change of the measured physical quantity
- **External noise unrelated is also easy to mix**

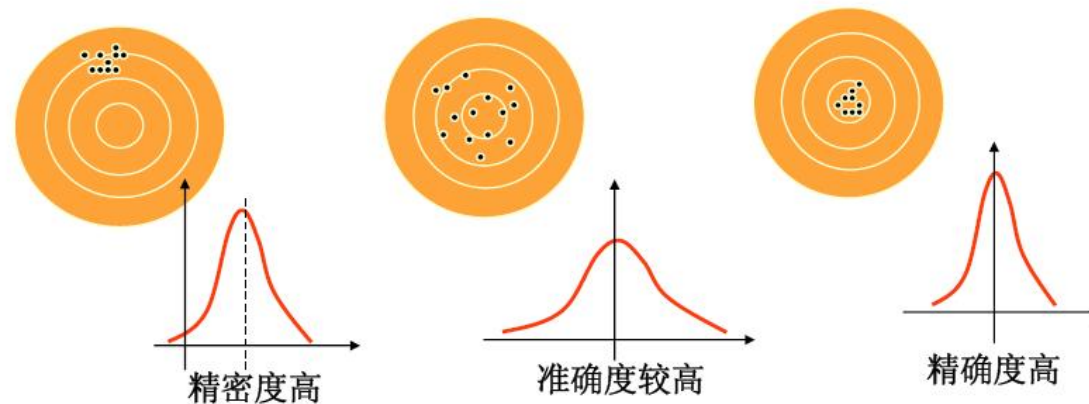


5.3.1 Characteristics of Sensors

Precision-related Index

- **Accuracy**: how close a measure value is to the true value
- **Precision**: how close measure values are to each other, basically how many decimal places are at the end of a given measurement.
- Higher the accuracy of sensor → More expensive
- Fulfill both the **accuracy**, **precision** and **financial** requirements

精度（精确度）、精密度与准确度



精度反映了测试系统中系统误差和随机误差的综合影响



5.3.1 Characteristics of Sensors

Dynamic Performance Index

- **Damping ratio, Time constant, Frequency response characteristic, Natural frequency, Critical velocity, etc.**
- **Frequency domain related measurement**



5.3.2 The Environment of Measurement

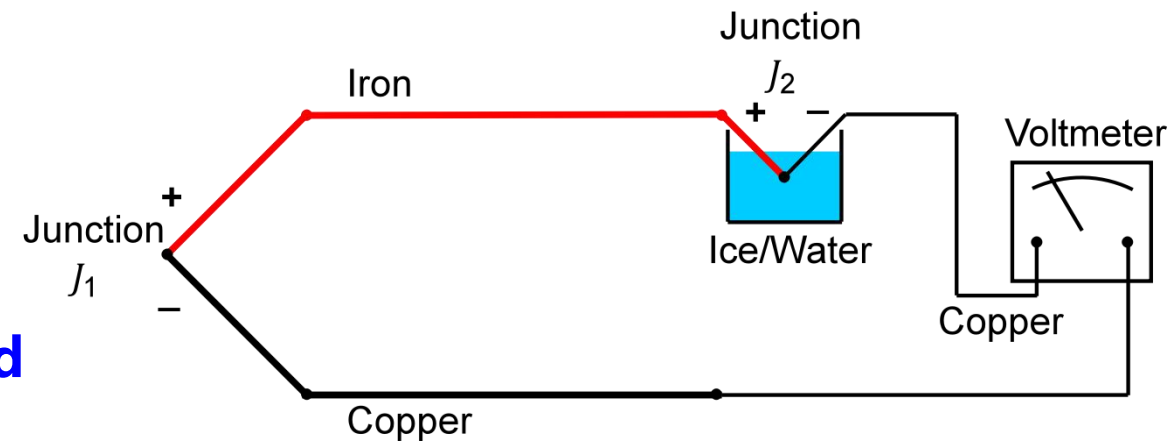
Environmental parameters is important on the selection of sensors

➤ Ambient Temperature

- Physical and chemical properties of the sensitive element will change with the temperature (temperature drift phenomenon)
- Selection of proper sensors to compensate the temperature effect

Example

- Thermoelectric potential is only related to the temperature of the **hot** junction, while the **cold** junction will fluctuate with the temperature





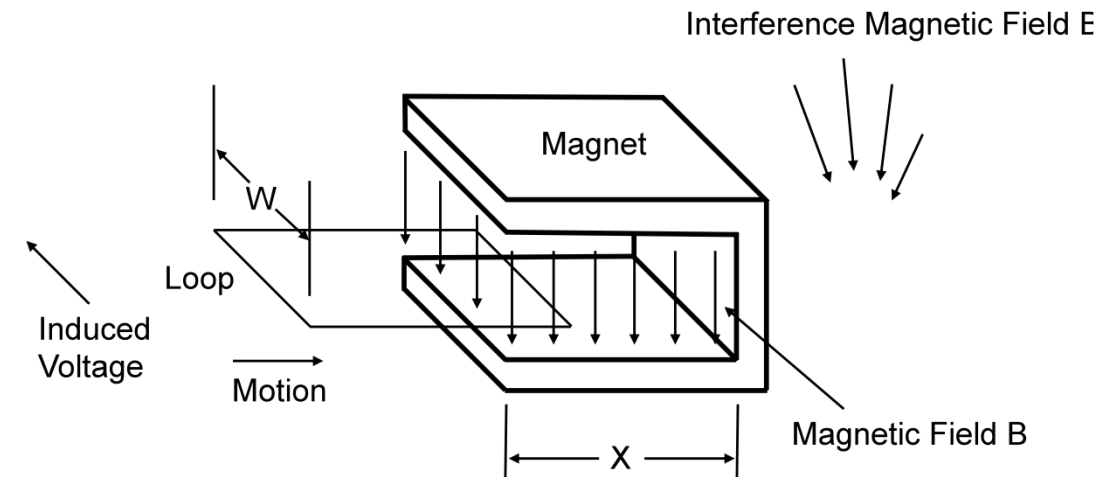
5.3.2 The Environment of Measurement

➤ Electromagnetic Interference

- For inductive sensors and magnetoelectric sensors, the electromagnetic field interference strongly disturbs the measurement

Example

- Eddy current sensor related to the magnetic flux passing through its metal conductors
- Magnetic field fluctuations change the magnetic flux of the environment, affecting the accuracy of the measurement
- **Interference shielding** is selected according to the electromagnetic field signals to reduce the noise





5.3.2 The Environment of Measurement

➤ Vibration and Impact

- For pressure sensors, vibration and impact in the environment affect measurements
- Vibration shock transmitted to the piezoelectric element → an additional current
- Strong vibration shock → damage the fragile components
- **Select the sensor that can withstand the vibration and impact**

➤ Other Environmental Factors

- Environmental humidity, corrosion degree of medium



5.3.3 Cost and Maintenance

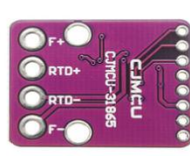
- Balance between cost and performance
- Maintenance: complexity of maintenance and convenience of procurement and replacement

Example: Temperature detection circuit

- If the temperature is -50°C - 250°C and there is no demand in linearity, NTC Thermistor is the best choice due to its low price.
- If the temperature is -200°C - 600°C and the system requires high stability and linearity, platinum RTD will be the best choice even though its price is the highest.



NTC Thermistor



Platinum RTD



Thermocouple



Semiconductor Based

Don't pay for more than what is actually needed

	NTC Thermistor [↵]	Platinum RTD [↵]	Thermocouple [↵]	Semiconductor Based [↵]
Temperature Range [↵]	-50 to 250°C [↵]	-200 to 600°C [↵]	-200 to 1750°C , depending upon type [↵]	-70 to 150°C [↵]
Stability [↵]	Epoxy Coated: [↵] $0.2^{\circ}\text{C}/\text{year}$ [↵] Hermetically Sealed: $0.02^{\circ}\text{C}/\text{year}$ [↵]	Film: $0.05^{\circ}\text{C}/\text{year}$ [↵] Full Wire: $0.002^{\circ}\text{C}/\text{year}$ [↵]	$>1^{\circ}\text{C}/\text{year}$ [↵]	$2^{\circ}\text{C}/\text{year}$ [↵]
Output [↵]	$-4.4\%/^{\circ}\text{C}$, flexible as output can be used as resistance or voltage [↵]	$0.00385\text{ Ohm}/^{\circ}\text{C}$ [↵]	10 to $40\text{ mV}/^{\circ}\text{C}$ [↵]	Various [↵]
Linearity [↵]	Exponential, Requires linearization [↵]	Fairly linear [↵]	Non-linear [↵]	Linear [↵]
Power Source [↵]	Any constant voltage or current [↵]	Any constant voltage or current [↵]	Self-powered [↵]	$4\text{-}30\text{V DC}$ [↵]
Typical Response Time [↵]	Fast: 0.12 to 10s [↵]	Very slow: 1 to 50s [↵]	Slow: 0.2 to 20s [↵]	Very slow: 5 to $60+\text{s}$ [↵]
Accuracy [↵]	0.05 to 1.5°C [↵]	0.1 to 1°C [↵]	0.5 to 5°C [↵]	1 to 5°C [↵]
Susceptibility to Electrical Noise [↵]	Very low [↵]	Very low [↵]	Extreme, especially if cold junction [↵]	Dependent upon board layout [↵]
Effect of lead resistance on Accuracy [↵]	Very low [↵]	Highly susceptible, especially for 3- and 4-wire configurations [↵]	None [↵]	None [↵]
Cost [↵]	Low to moderate, Hermetically sealed [↵]	High [↵]	Medium [↵]	Medium [↵]



Outline

5.4

Applications and Development of Sensor

- 5.4.1 Main Applications
- 5.4.2 Development Trends



5.4 Applications and Development Trends of Sensor

➤ 5.4.1 Main Applications

- Security monitoring, Medical and health, Automatic control and Industrial production.

Security Monitoring

- **Infrared human body detectors:** anti-theft
- Methane sensors: gas leakage
- PM2.5 sensors: air pollution
- Thermostat with a temperature sensor, a smoke sensor and a brightness sensor: building security
- Photoelectric displacement sensor in the mouse
- CCD sensors in digital cameras
- Fingerprint and tactile sensors in smart phones
-

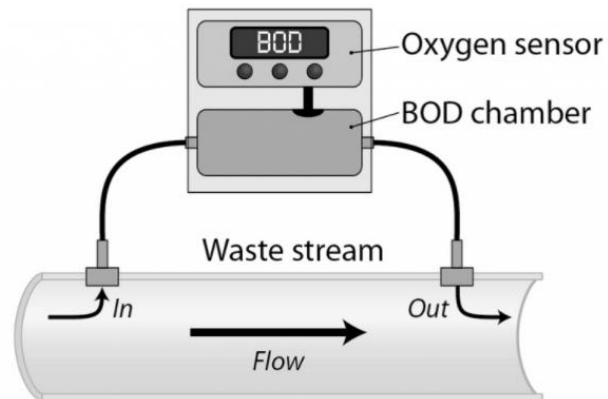
Infrared Human Body Detector



5.4.1 Main Applications

Medical and Health

- Biosensors use living microorganisms as sensitive materials
- Glucose sensors: blood glucose concentration of diabetic patients
- Biological oxygen demand (BOD) sensors: types of bacteria at the infection site
- Enzyme sensors: detect the concentration of pesticides



BOD Sensor



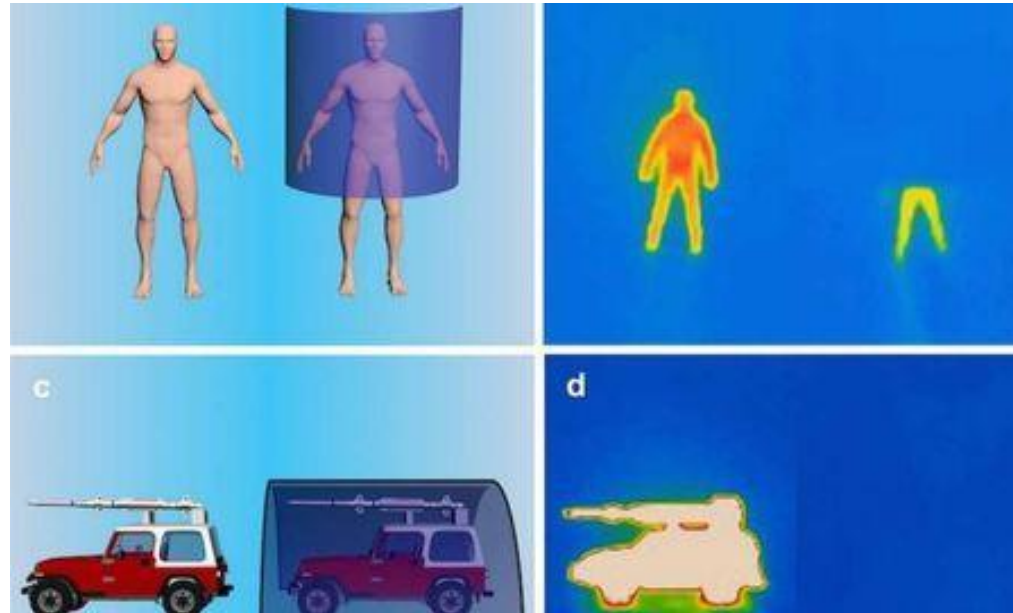
Glucose Sensor

COVID-19 virus detector

5.4.1 Main Applications

Military Technology

- Objects of different temperatures have different wavelengths of the released infrared energy
- **Infrared sensing technology** can penetrate smoke, fog, rain and other obstacles
- Night vision combat instruments, infrared guidance, target recognition, and photoelectric countermeasures

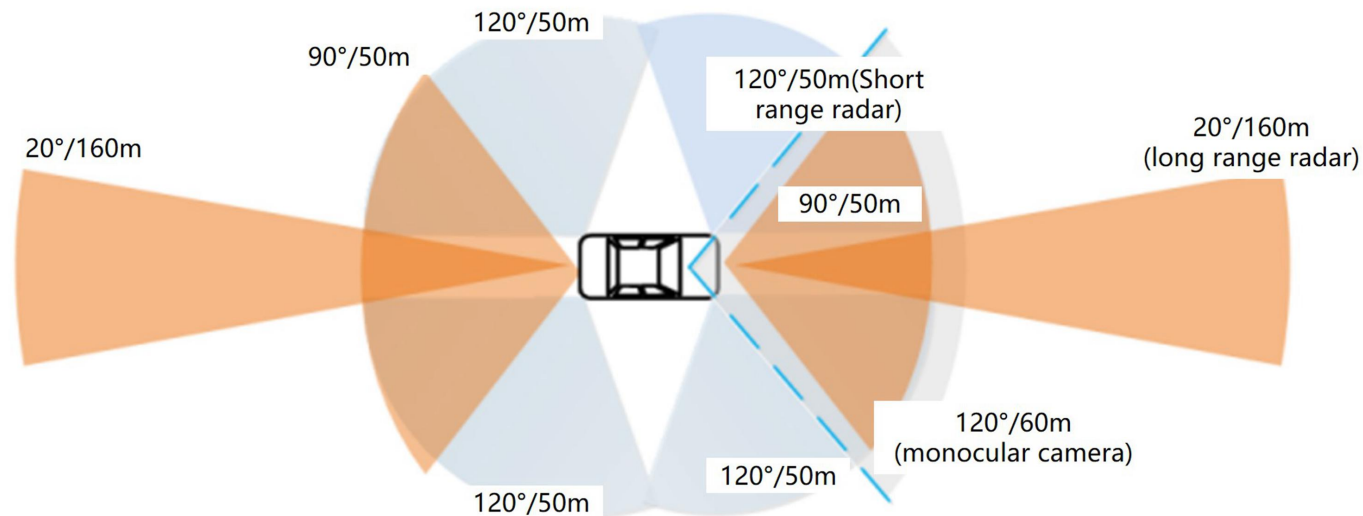


Infrared Sensing Technology

5.4.1 Main Applications

Industrial Production and Automatic Control

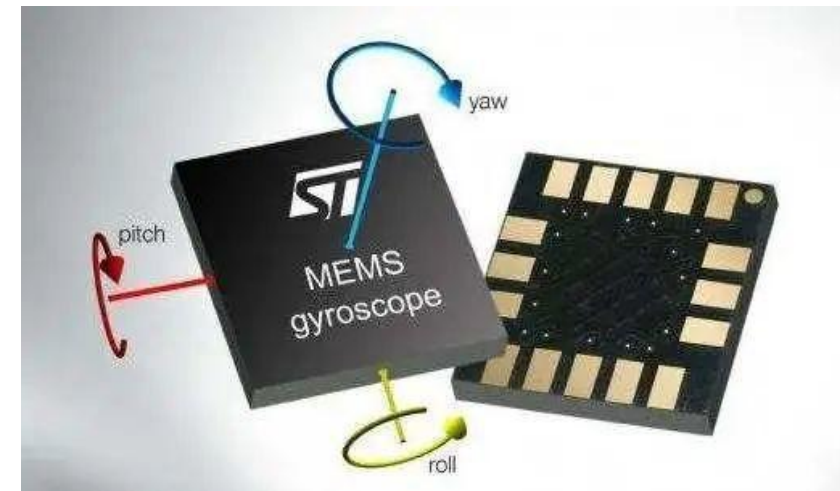
- In the production of petroleum, chemical industry, electric power, iron and steel, machinery, etc., it is necessary to detect the information of various process parameters in time, and automatically control the production process through electronic computers or controllers
- **Self-driving vehicle:** distance sensors, pressure sensors, speed sensors, temperature sensors, position sensors, and collision sensors



5.4.2 Development Trends

➤ Development of New-type Sensor

- Magnetic sensors using quantum mechanical effect: detect extremely weak magnetic field strength
- New sensitive materials: ceramic materials, optical fiber materials, superconducting materials, organic materials, composite materials, nanomaterials, smart materials
- MEMS (Micro-Electro-Mechanical System) sensors: small size, light weight, low energy consumption, high reliability, low cost, easy batch and integration



MEMS Sensors



5.4.2 Development Trends

➤ Integration and Multi-function of Sensor

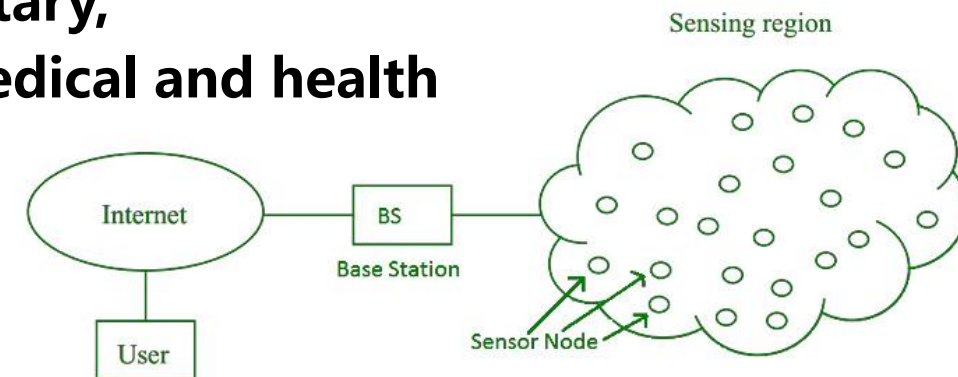
- Micro-machining technology can integrate sensitive materials and circuits on a chip and integrate the sensors with subsequent conditioning and compensation circuits
- **Signal** conversion function → **Multiple** functions (amplification, calculation and interference compensation)
- Multi-functional sensors integrate sensitive components such as temperature, force, and heat on the same chip
- Realize multiple functions at the same time with a single sensor system



5.4.2 Development Trends

➤ Smart and Networked Sensor

- Traditional sensor and a microprocessor
- One or more sensitive functions
- Combination of sensors and microprocessors → Detection & Information processing, logical judgment, self-diagnosis and thinking
- **Networked sensors**: Randomly distributed homogeneous or heterogeneous sensor nodes
- **Wireless sensor network**: national defense and military, environmental monitoring, traffic management, medical and health





Summary

- **Sensor component: Sensitive components, conversion components, and signal conditioning circuits**
- **Basic characteristics of sensor: static characteristic, dynamic characteristic and other performances**
- **Fundamental concepts of sensor calibration**
- **Selection of a proper sensor in terms of the specific measurement purpose**
- **Sensor characteristics: range, sensitivity and precision**
- **Environmental factors (temperature, humidity and vibration) and financial factors (cost and maintenance)**
- **Applications and development trends of sensor technology**